



We are on a mission to address the digital skills gap for 10 Million+ young professionals, train and empower them to forge a career path into future tech

Javascript Insights – Java Script Execution for Selenium

MARCH, 2026



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- Importance of using Javascript in Selenium
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What is Javascript Executor?



What is Javascript Executor?

What is JavaScript Executor?

- JavaScript Executor is an Interface that helps to execute JavaScript through Selenium Web driver.
- JavaScript Executor provides two methods "execute_script" & "execute_async_script" to run JavaScript on the selected window or current page.



Importance of using Javascript in Selenium



Importance of using Javascript in Selenium

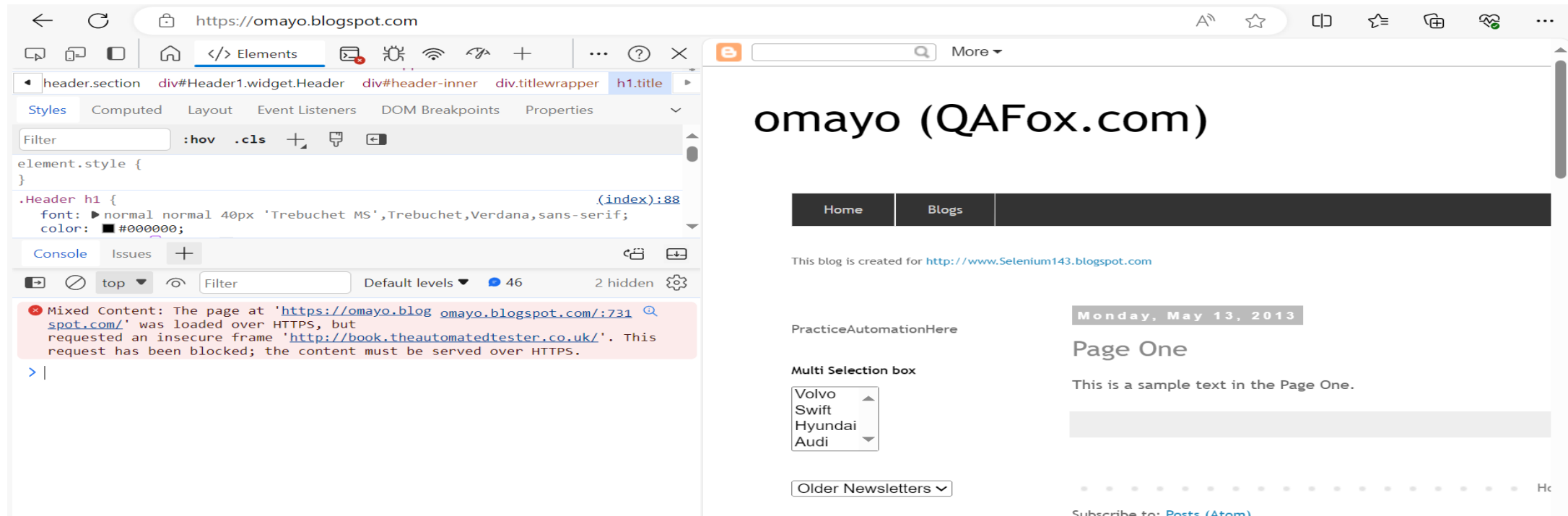
Importance of using Java script in Selenium

- In Selenium Web driver, locators like XPath, CSS, etc. are used to identify and perform operations on a web page
- In case, these locators do not work you can use JavaScript Executor. JavaScript Executor is used to perform a desired operation on a web element
- It provides advantages over Find Element method while handling tricky XPath as well as finding element which are sometimes hidden
- You can directly interact and alter elements on the page

Handling Web Elements using javascript executor

Handling alert using javascript executor

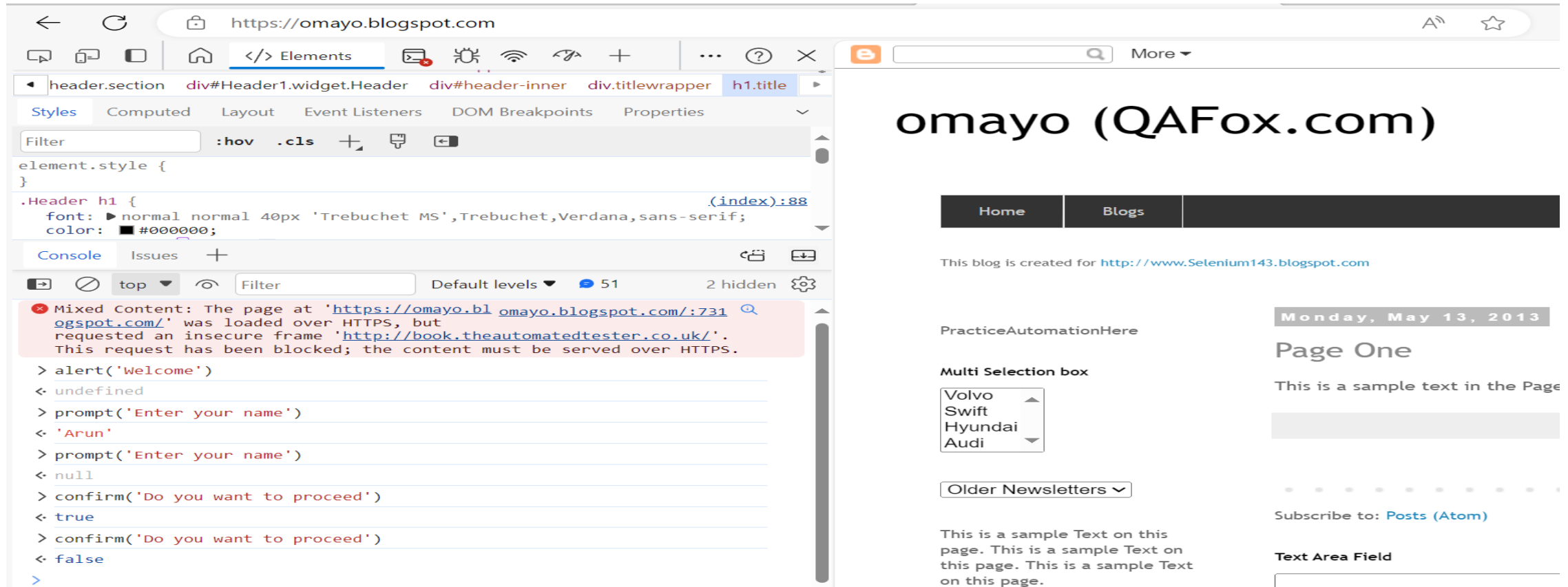
- Open the web application in the web browser and right click and choose inspect.
- Inspect window will open, choose console



Handling Web Elements using javascript executor

Handling alert using javascript executor

- JavaScript code to handle and display various alerts in web page are given in the console:



The screenshot shows a web browser window with the URL `https://omayo.blogspot.com`. The browser's developer tools are open, displaying the **Console** tab. The console shows a series of JavaScript execution results:

```

> alert('Welcome')
< undefined
> prompt('Enter your name')
< 'Arun'
> prompt('Enter your name')
< null
> confirm('Do you want to proceed')
< true
> confirm('Do you want to proceed')
< false
>

```

A red error message is visible at the top of the console:

```

Mixed Content: The page at 'https://omayo.blogspot.com/' was loaded over HTTPS, but requested an insecure frame 'http://book.theautomatedtester.co.uk/'. This request has been blocked; the content must be served over HTTPS.

```

The web page content is visible on the right side of the screenshot. It has a title **omayo (QA Fox.com)** and a navigation bar with links for **Home** and **Blogs**. Below the navigation bar, there is a message: "This blog is created for <http://www.Selenium143.blogspot.com>".

The main content area includes a section titled **PracticeAutomationHere** with a sub-section **Multi Selection box**. This section contains a multi-select dropdown menu with options: Volvo, Swift, Hyundai, and Audi. Below this is a section for **Older Newsletters** with a dropdown arrow.

On the right side of the page, there is a date stamp: **Monday, May 13, 2013**, followed by the text **Page One** and a paragraph: "This is a sample text in the Page". Below this is a text area field labeled **Text Area Field**.

Alerts – Javascript - Program 1

```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()

driver.get("https://omayo.blogspot.com/")

driver.execute_script("alert('Welcome')")
time.sleep(4)
driver.execute_script("prompt('Enter your name')")
time.sleep(4)
driver.execute_script("confirm('Do you want to proceed')")
time.sleep(4)
```

Clicking an Element using Javascript Executor- Program 2

```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()

driver.get("https://omayo.blogspot.com/")
#Method 1
#driver.execute_script("document.getElementById('alert1').click()")

#Method2
button = driver.find_element(By.ID, value='alert1')
driver.execute_script("arguments[0].click()",button)
time.sleep(4)
```


Clicking an Element using Javascript Executor- Program 2

```
#Reusable Javascript Click Function
```

```
def click_elefunction(element):  
    driver.execute_script("arguments[0].click()",element)
```

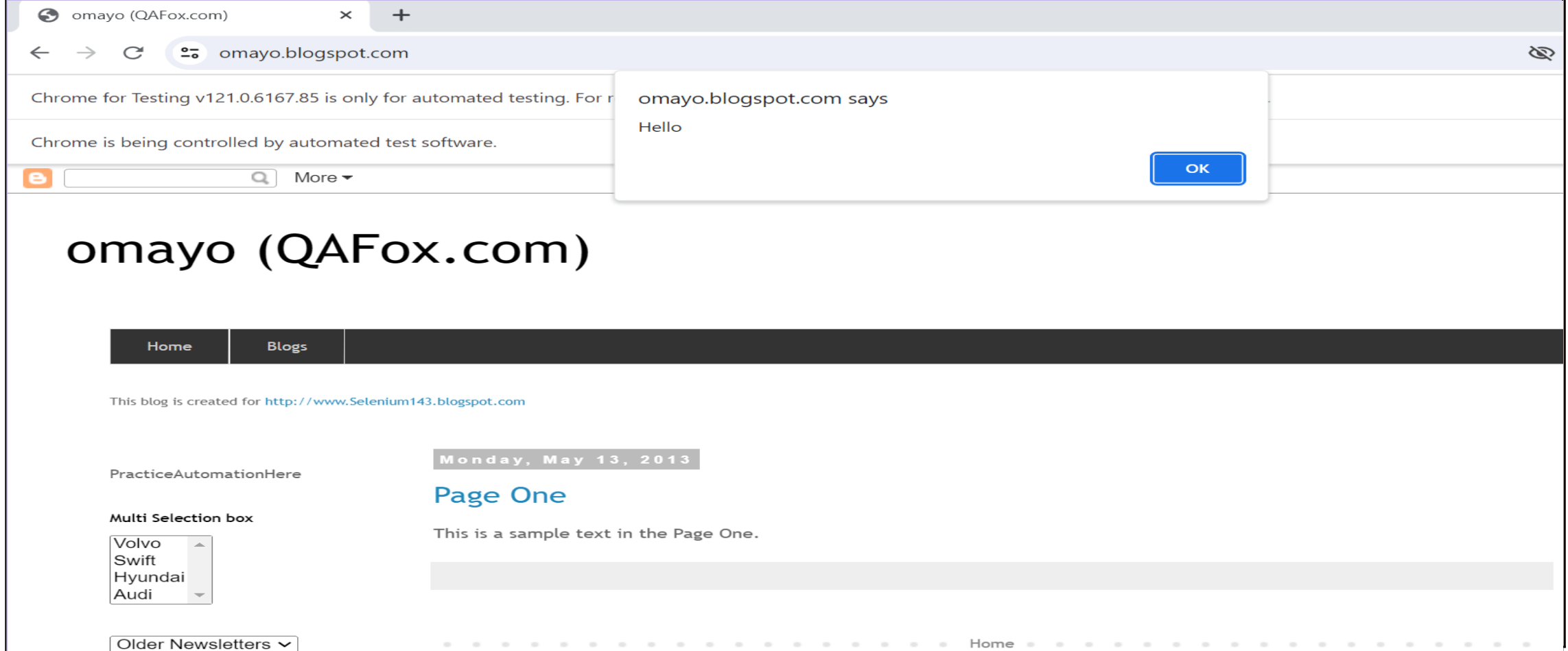
```
#Reusable Click Function call
```

```
click_elefunction(button)  
time.sleep(4)
```

Why do we need selenium waits?

An Example Scenario

Output:



Flashing an Element using Javascript Executor- Program 3

```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()

# Flashing an element
driver.get("https://tutorialsninja.com/demo/index.php?route=account/login")
```

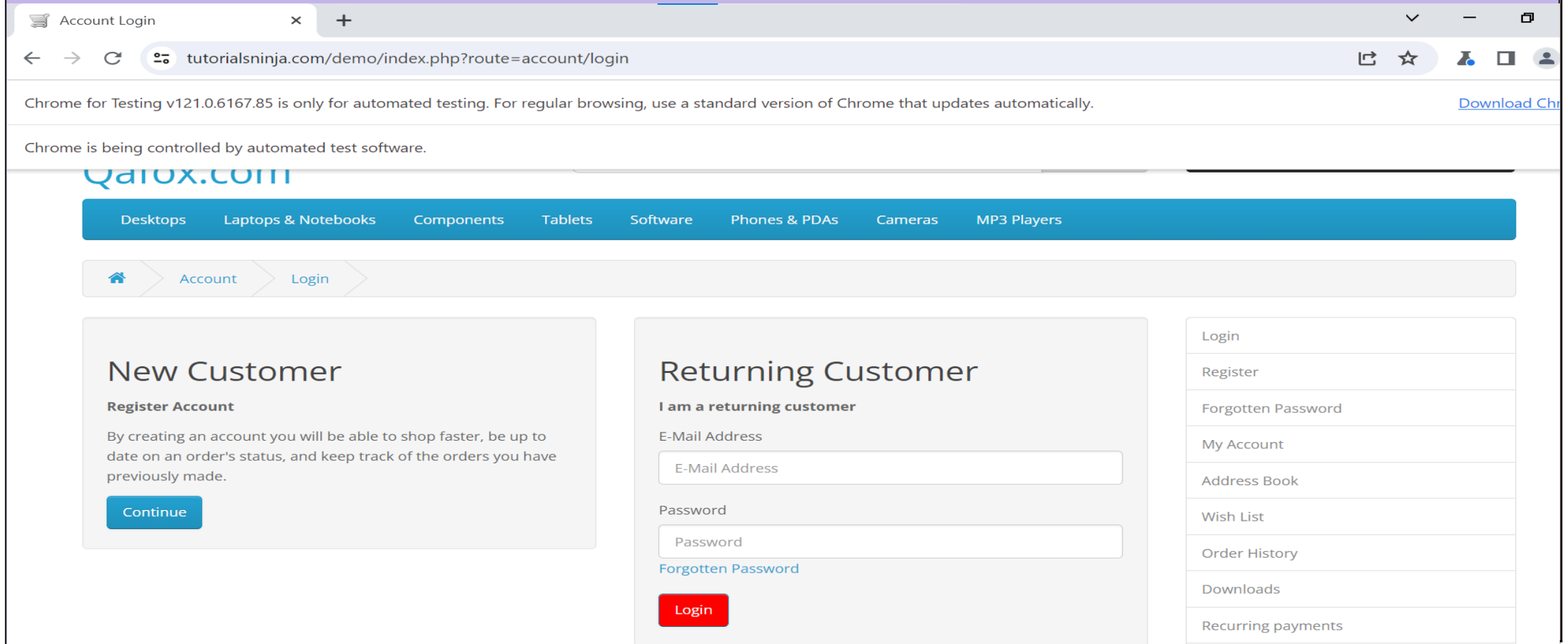
Flashing an Element using Javascript Executor- Program 3

```
def flash_element (element):  
    for i in range(1, 30):  
        driver.execute_script("arguments[0].style.background='red'",element)  
        time.sleep(.2)  
        default_color = element.value_of_css_property('color')  
        driver.execute_script("arguments[0].style.background='"+default_color+"'",element)  
  
login_button = driver.find_element(By.XPATH, value="//input[@type='submit']")  
  
flash_element(login_button)  
time.sleep(4)
```


Why do we need selenium waits?

An Example Scenario

Output:



The screenshot shows a web browser window with the title "Account Login". The address bar displays the URL: `tutorialsninja.com/demo/index.php?route=account/login`. Below the browser window, there is a message: "Chrome for Testing v121.0.6167.85 is only for automated testing. For regular browsing, use a standard version of Chrome that updates automatically." and another message: "Chrome is being controlled by automated test software."

The web application interface includes a navigation bar with the following categories: Desktops, Laptops & Notebooks, Components, Tablets, Software, Phones & PDAs, Cameras, and MP3 Players. Below the navigation bar, there is a breadcrumb trail: Home > Account > Login.

The main content area is divided into two columns. The left column is titled "New Customer" and contains the sub-header "Register Account". It includes a paragraph: "By creating an account you will be able to shop faster, be up to date on an order's status, and keep track of the orders you have previously made." and a blue "Continue" button.

The right column is titled "Returning Customer" and contains the sub-header "I am a returning customer". It includes two input fields: "E-Mail Address" and "Password". Below the "Password" field, there is a link for "Forgotten Password". At the bottom of this column is a red "Login" button.

On the far right, there is a vertical menu with the following links: Login, Register, Forgotten Password, My Account, Address Book, Wish List, Order History, Downloads, and Recurring payments.

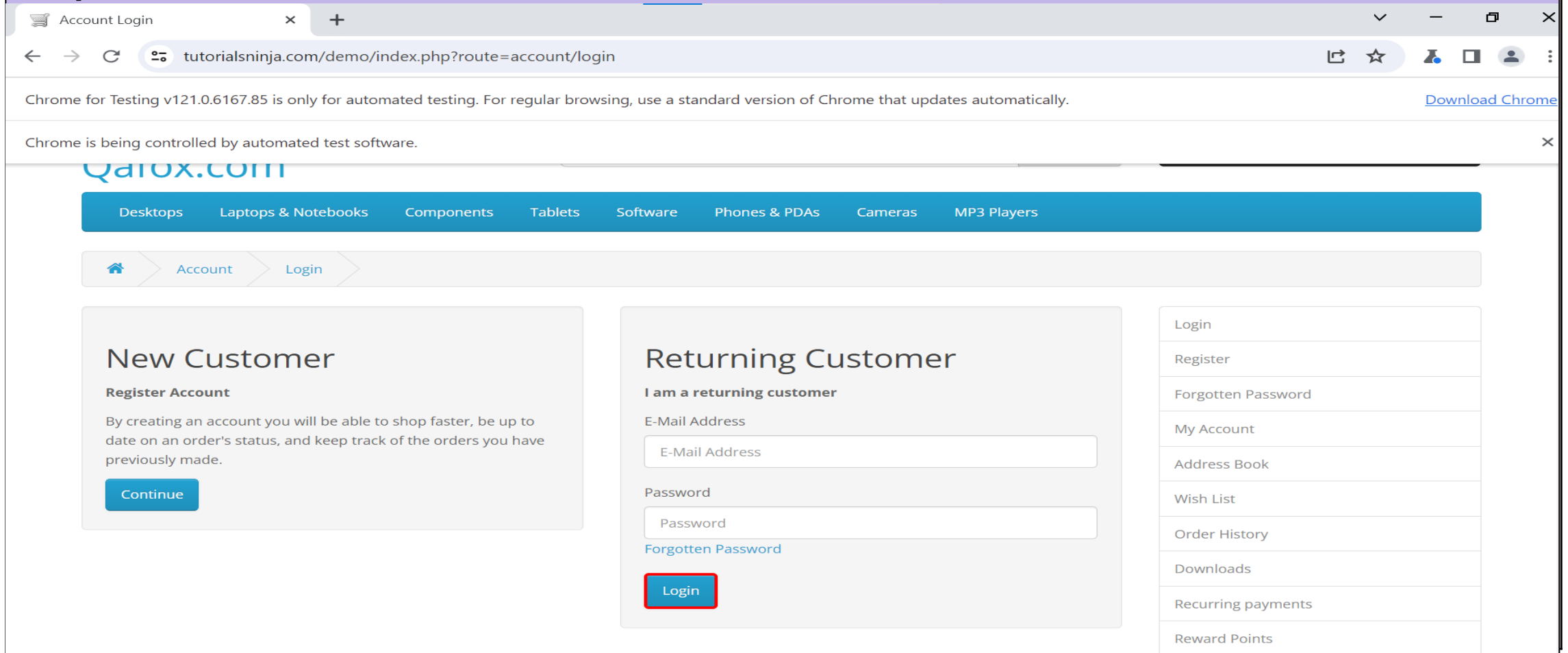
Bordering an Element using Javascript Executor- Program 4

```
def element_border(element):  
    driver.execute_script("arguments[0].style.border=`4px solid red\"",element)  
  
login_button = driver.find_element(By.XPATH, value="//input[@type='submit']")  
  
element_border(login_button)  
time.sleep(5)
```

Why do we need selenium waits?

An Example Scenario

Output:



Account Login

tutorialsninja.com/demo/index.php?route=account/login

Chrome for Testing v121.0.6167.85 is only for automated testing. For regular browsing, use a standard version of Chrome that updates automatically. [Download Chrome](#)

Chrome is being controlled by automated test software.

Qatoox.com

Desktops Laptops & Notebooks Components Tablets Software Phones & PDAs Cameras MP3 Players

Home Account Login

New Customer

Register Account

By creating an account you will be able to shop faster, be up to date on an order's status, and keep track of the orders you have previously made.

[Continue](#)

Returning Customer

I am a returning customer

E-Mail Address

Password

[Forgotten Password](#)

[Login](#)

Login

Register

Forgotten Password

My Account

Address Book

Wish List

Order History

Downloads

Recurring payments

Reward Points

Retrieving title of a page using Javascript Executor- Program 5

```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()
driver.get("https://www.google.co.in")

page_title = str(driver.execute_script("return document.title"))
print(page_title)
page_url = str(driver.execute_script("return document.URL"))
print(page_url)

driver.quit()
```

Why do we need selenium waits?

An Example Scenario

Output:

PROBLEMS

26

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Code



```
[Running] python -u "c:\arunproject\VSCodePython\java_script.py"
```

Google

<https://www.google.co.in/>

```
[Done] exited with code=0 in 13.425 seconds
```


Entering text into Text Fields using Javascript Executor- Program 6

```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()

def type_text(element, input_text):
    driver.execute_script("arguments[0].value='"+input_text+"'", element)

driver.get("https://tutorialsninja.com/demo/index.php?route=account/login")

user_name = driver.find_element(By.XPATH, value="//input[@name='email']")

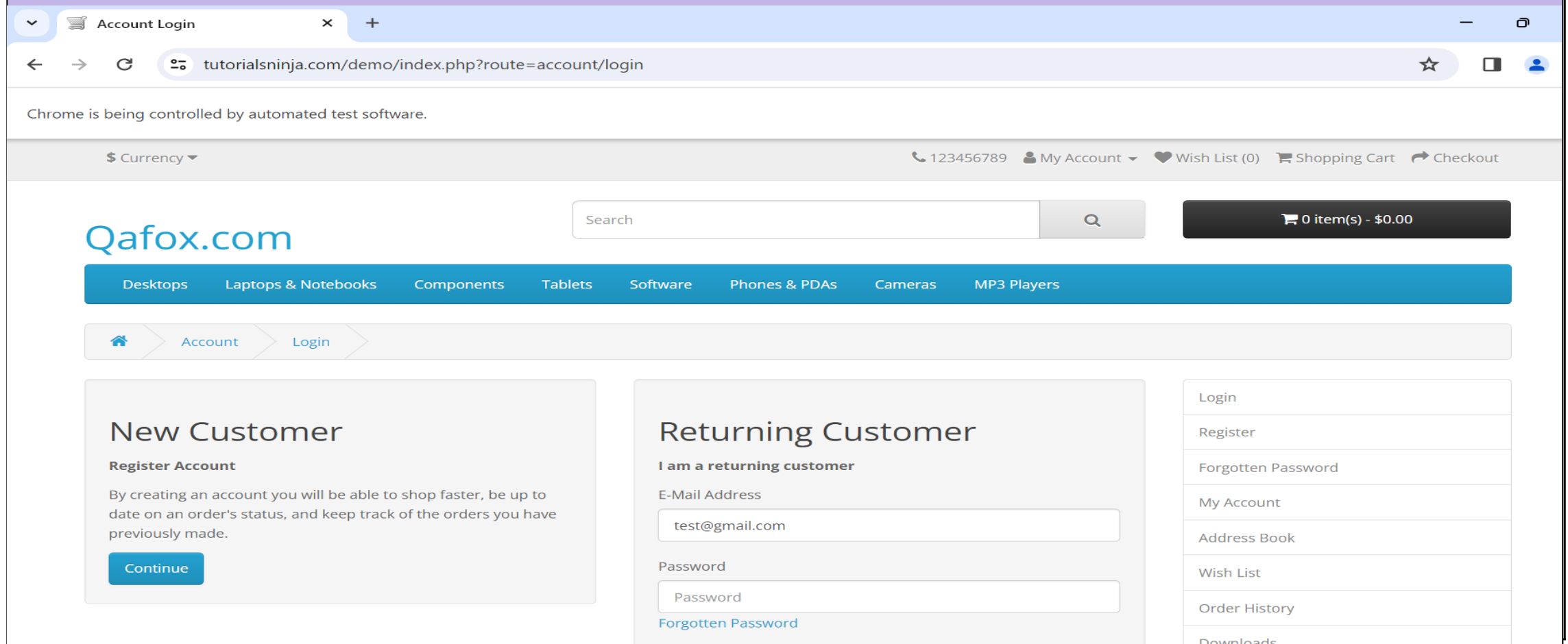
type_text(user_name, 'test@gmail.com')

driver.quit()
```

Why do we need selenium waits?

An Example Scenario

Output:



Account Login

tutorialsninja.com/demo/index.php?route=account/login

Chrome is being controlled by automated test software.

\$ Currency

123456789 My Account Wish List (0) Shopping Cart Checkout

Search

0 item(s) - \$0.00

Desktops Laptops & Notebooks Components Tablets Software Phones & PDAs Cameras MP3 Players

Home Account Login

New Customer

Register Account

By creating an account you will be able to shop faster, be up to date on an order's status, and keep track of the orders you have previously made.

Continue

Returning Customer

I am a returning customer

E-Mail Address

test@gmail.com

Password

Password

[Forgotten Password](#)

- Login
- Register
- Forgotten Password
- My Account
- Address Book
- Wish List
- Order History
- Downloads

Selecting a date in calendar - Program 7

```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()

def type_date(element, date):
    driver.execute_script("arguments[0].value='"+date+"'", element)

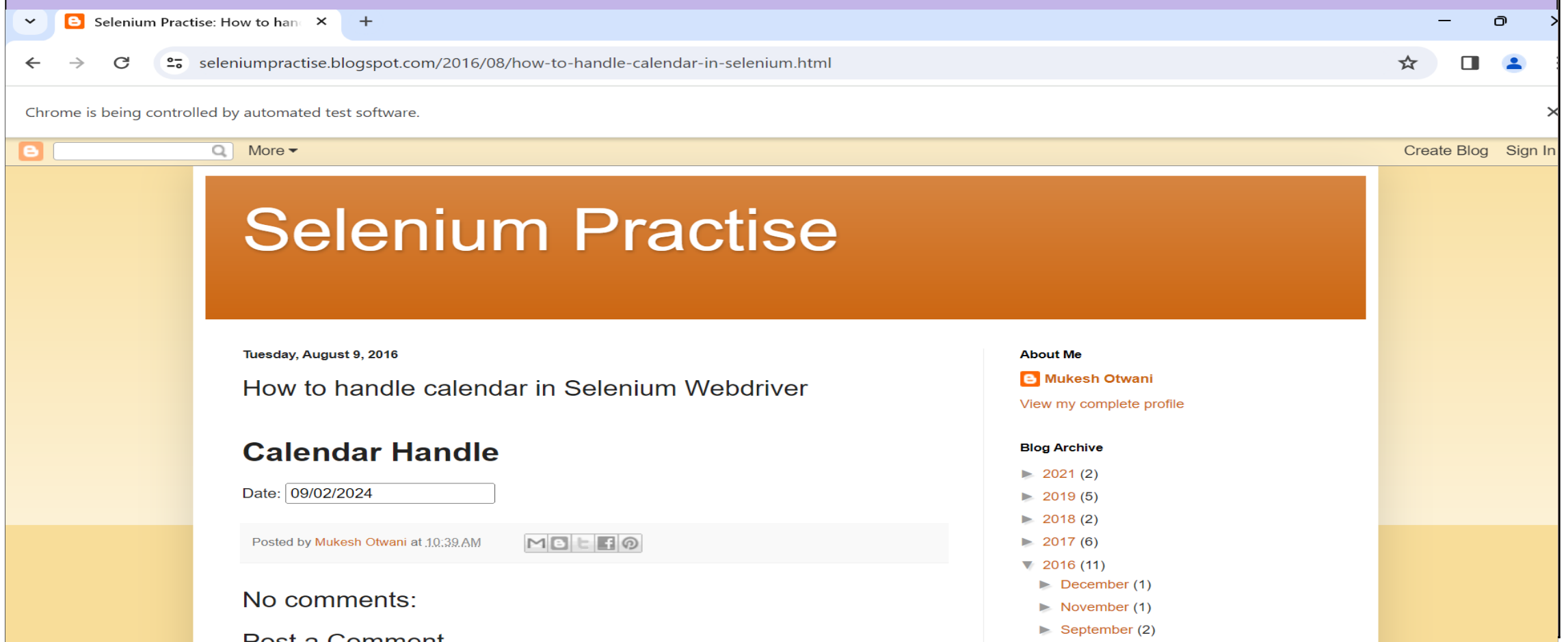
driver.get("https://seleniumpractise.blogspot.com/2016/08/how-to-handle-calendar-in-selenium.html")
calendar = driver.find_element(By.ID, value="datepicker")
type_date(calendar, '09/02/2024')

driver.quit()
```

Why do we need selenium waits?

An Example Scenario

Output:



The screenshot shows a web browser window with the following details:

- Tab:** Selenium Practise: How to han...
- Address Bar:** seleniumpractise.blogspot.com/2016/08/how-to-handle-calendar-in-selenium.html
- Message:** Chrome is being controlled by automated test software.
- Page Header:** Search bar, More ▾, Create Blog, Sign In
- Main Content:**
 - # Selenium Practise
 - Tuesday, August 9, 2016
 - ## How to handle calendar in Selenium Webdriver
 - ### Calendar Handle
 - Date:
 - Posted by **Mukesh Otwani** at 10:39 AM
 - No comments:
 - Post a Comment
- Right Sidebar:**
 - About Me**
 -  **Mukesh Otwani**
 - [View my complete profile](#)
 - Blog Archive**
 - ▶ 2021 (2)
 - ▶ 2019 (5)
 - ▶ 2018 (2)
 - ▶ 2017 (6)
 - ▼ 2016 (11)
 - ▶ December (1)
 - ▶ November (1)
 - ▶ September (2)

Refreshing, scrolling through and scrapping text - Program 8

```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome()
driver.maximize_window()
driver.get("https://omayo.blogspot.com/")
driver.execute_script("history.go(0)")
time.sleep(5)
driver.execute_script("window.scrollTo(0,500)")
time.sleep(5)
driver.execute_script("history.go(0)")
time.sleep(5)
driver.execute_script("window.scrollTo(0,document.body.scrollHeight)")
time.sleep(5)
driver.execute_script("window.scrollTo(0,-500)")
time.sleep(5)
```

Refreshing, scrolling through and scrapping text - Program 8

```
blogger = driver.find_element(By.XPATH, value="//a[text()='Blogger']")
driver.execute_script("arguments[0].scrollIntoView(true);", blogger)

page_text = str(driver.execute_script("return document.documentElement.innerText"))
print(page_text)

driver.quit()
```

Why do we need selenium waits?

An Example Scenario

Output:

```
PROBLEMS 26 OUTPUT DEBUG CONSOLE TERMINAL PORTS Code
[Done] exited with code=0 in 10.137 seconds

[Running] python -u "c:\arunproject\VSCodePython\java_script.py"
omayo (QAfox.com)

Home
Blogs

This blog is created for http://www.Selenium143.blogspot.com

Monday, May 13, 2013
Page One
This is a sample text in the Page One.
Home
Subscribe to: Posts (Atom)
Text Area Field
Text Area Field Two
Table
Name    Age Place
Kishore 22  Delhi
Manish  25  Pune
Praveen 29  Bangalore
Dheepthi 31  Mumbai
HTML Form
Username:

Password:
```


THANK YOU